

Assignment 13

Q1

1) [0.6 points]

Consider the following Turing machine M that recognises odd strings over $\Sigma = \{0,1\}$, $Q = \{q_0, q_1, q_{acc}, q_{rej}\}$, $\Gamma = \{0,1, \perp\}$, and the transition function:

$$\delta(q_0, 0) = \{q_0, 0, R\} \quad \delta(q_1, 0) = \{q_1, 0, R\}$$

$$\delta(q_0, 1) = \{q_1, 1, R\} \quad \delta(q_1, 1) = \{q_0, 1, R\}$$

$$\delta(q_0, \perp) = \{q_{rej}, \perp, R\} \quad \delta(q_1, \perp) = \{q_{acc}, \perp, R\}$$

Do the following:

- Construct the tableau of size 5 X 5 representing the only possible branch computation of M for the string $w = 111$
- Write down the Boolean formulas ϕ_{start} and ϕ_{accept} of the polynomial time reduction from the language of odd strings over $\Sigma = \{0,1\}$ and SAT.

A1

a.

q_0	1	1	1	$\perp$
1	q_1	1	1	$\perp$
1	1	q_0	1	$\perp$
1	1	1	q_1	$\perp$
1	1	1	$\perp$	q_{acc}

b.

$$\phi_{start} = x_{(1,1,q_0)} \wedge x_{(1,2,w_1)} \wedge x_{(1,3,w_2)} \wedge \dots \wedge x_{(1,n+1,w_n)} \wedge x_{(1,n+2,\perp)} \wedge \dots \wedge x_{(1,n^k,\perp)}$$

$$\phi_{accept} = \bigvee_{1 \leq j \leq n^k} x_{(n^k,j,q_{accept})}$$

Q2

2) [0.4 points]

Say which of the following windows are legal, hence may appear in a tableau representing a branch of computation of a nondeterministic Turing machine:

a)

b	q_1	b
q_2	b	q_2

b).

a	b	a
a	a	a

c)

a	q_1	b
q_2	a	c

d)

a	a	q_1
a	a	b

e)

a	q_1	b
a	a	q_2

b	b	b
c	b	b

a	b	a
a	b	q_2

f)

g)

A2

- (a) Illegal.
There cannot be two states in the same row.
- (b) Illegal.
The “b” in the first row is not adjacent to a state, which means it cannot be revised by state change.
- (c) Legal.
By state change, the pointed data by the state is changed from “b” to “c”.
- (d) Legal.
Assume the nearest right 2-row column adjacent to the block is “b” and “ q_2 ” from top to bottom, then it’s legal as the state change does not cause data change.
- (e) Legal.
It’s legal when state changes from “ q_1 ” to “ q_2 ”, with pointed data changing from “b” to “a”.
- (f) Legal.
It can be a legal part of the following legal figure.

a	q_1	b	b	b
q_2	a	c	b	b

- (g) Legal.
It can be a legal part of the following legal figure.

a	b	a	q_1
a	b	q_2	a

Q3

3) [0.6 points]

In the proof of Cook-Levin Theorem, a window is a 2 X 3 rectangle of cells. Show why the proof would have failed if we had used a 2 X 2 rectangle instead.

A3

If we use the same proof of Cook-Levin Theorem, but using 2×2 windows instead of 2×3 . We have to identify $\emptyset_{\text{Move}}$ is true or not by identifying if all the 2×2 windows are legal or not.

Assume we have a rule to identify a 2×2 window is legal or not.

Obviously, if the upper row and the lower row has the same symbols, it is legal.

For $\delta(q, a) = (q', a, L)$, $\delta(q, b) = (q', b, R)$

The following move are both legal,

...	a	q	b	a	...
...	a	b	q'	a	...

...	a	b	q	a	...
...	a	q'	b	a	...

The "..." means the upper row and lower row have the same symbols.

Then, the following windows should be all legal:

a	q
a	b

q	a
b	a

Then the following move should also be legal,

...	a	q	a	...
...	a	b	a	...

But it obviously is illegal.

In this case, there can not exist such a rule that can allow us to use 2×2 windows to prove the Theorem in the same way.

Q4

4) [1.5 point]

Let $UHAMPATH = \{ \langle G, s, t \rangle \mid G \text{ is an undirected graph with a Hamiltonian path from } s \text{ to } t \}$.

Show that $UHAMPATH$ is NP-Complete using a reduction from $HAMPATH$.

HINT: Except for s and t you will want to replace each node u of the original directed graph G with 3 nodes in G' : u^{in}, u^{mid}, u^{out}

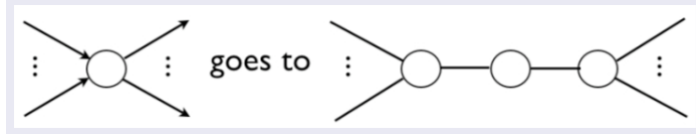
A4

- We reduce $HAMPATH$ to $UHAMPATH$.
- All nodes except s and t in the directed graph G , map to 3 nodes in the undirected graph G' .
- G has a Hamiltonian path $\Leftrightarrow G'$ has an undirected Hamiltonian path.

Construction:

- s in G maps to s^{out} in G' .
- t in G maps to t^{in} in G' .
- Any other node u_i maps to $u_i^{in}, u_i^{mid}, u_i^{out}$ in G' .

- All arcs coming to u_i in G become edges incident on u_i^{in} in G' .
- All arcs going out from u_i in G become edges incident on u_i^{out} in G' .



Correctness:

- Note that if

$$s, u_1, u_2, \dots, u_k, t$$

is a Hamiltonian path in G , then

$$s^{out}, u_1^{in}, u_1^{mid}, u_1^{out}, u_2^{in}, u_2^{mid}, u_2^{out}, \dots, u_k^{out}, t^{in}$$

is a Hamiltonian path in G' .

- Any Hamiltonian path between s^{out} and t^{in} must go through the triple of nodes except for the start and end nodes.

Q5

5) [1.5 point]

VERTEX COVER = $\{ \langle G, k \rangle \mid G \text{ is an undirected graph that has a } k - \text{node vertex cover} \}$. (A vertex cover is a subset of nodes such that every edge of the graph G touches one of the nodes).

Prove that the language is NP-Complete using a reduction from 3SAT.

HINT: You will need two kinds of gadget:

- Variable gadget: for each variable x_i you want that either it's in the cover ($x_i = \text{TRUE}$) or it isn't ($\bar{x}_i = \text{TRUE}$) – create a node for each and connect them so one of them must appear in the produced cover
- Clause gadget: you need to make sure that at least one variable in each clause is true – create a 3-clique with the variables of each clause and connect them to the corresponding variable gadget nodes
- With a correct construction you should end up with a cover of size $k = v + 2c$ where v is the number of variables in the 3SAT formula and c is the number of clauses

A5

We know that $VERTEX COVER \in NP$. Now we prove that it is NP-complete. Let

$$\phi = C_1 \wedge C_2 \wedge \dots \wedge C_m$$

be a 3CNF formula, where each clause has three literals.

We construct a graph G as follows.

For each clause $C_i = (a_i \vee b_i \vee c_i)$, create three vertices corresponding to its three literals:

$$a_i, b_i, c_i.$$

Connect these three vertices to form a triangle.

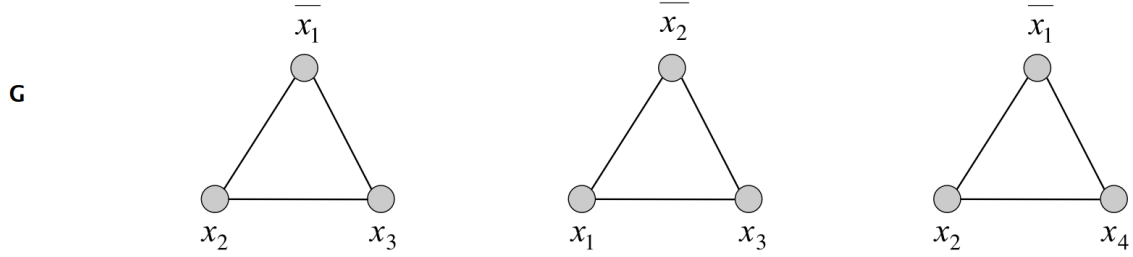
Next, for any two vertices corresponding to complementary literals, such as x and $\bar{x}$, add an edge between them.

The output of the reduction will be,

$$\langle G, 2m \rangle.$$

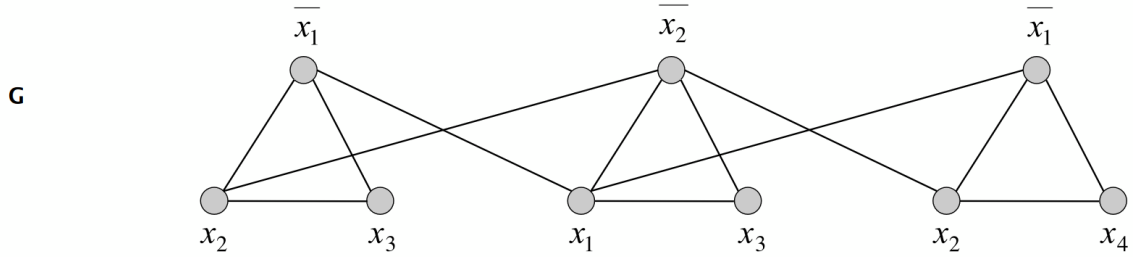
We now prove that,

$$\phi \in 3SAT \iff G \text{ has a vertex cover of size } 2m.$$



k = 3

$$\Phi = (\bar{x}_1 \vee x_2 \vee x_3) \wedge (x_1 \vee \bar{x}_2 \vee x_3) \wedge (\bar{x}_1 \vee x_2 \vee x_4)$$



k = 3

$$\Phi = (\bar{x}_1 \vee x_2 \vee x_3) \wedge (x_1 \vee \bar{x}_2 \vee x_3) \wedge (\bar{x}_1 \vee x_2 \vee x_4)$$

($\Rightarrow$)

Suppose ϕ is satisfiable. Let α be a satisfying assignment.

Since every clause is satisfied, each clause contains at least one true literal. For each clause triangle, choose one true literal and exclude its vertex from the vertex cover. Put the other two vertices of the triangle into the vertex cover.

Since there are m triangles and we choose 2 vertices from each triangle, the size of the set is $2m$.

We show that this set is a vertex cover.

First, every edge inside a clause triangle is covered, because in each triangle we include two of the three vertices. Any edge in a triangle must touch at least one of these two included vertices.

Second, consider an edge connecting two complementary literals, such as x and $\bar{x}$. These two literals cannot both be true under the same truth assignment. Since we only exclude vertices corresponding to chosen true literals, the two endpoints of this complementary edge cannot both be excluded. Therefore, at least one endpoint of the edge is included in the vertex cover.

Hence every edge is covered. Therefore, G has a vertex cover of size $2m$.

($\Leftarrow$)

Suppose G has a vertex cover S of size $2m$.

Each clause triangle needs at least two vertices in any vertex cover, because a triangle cannot be covered by only one vertex. So, S contains exactly two vertices from each clause triangle.

Therefore, in each triangle, exactly one vertex is not in S .

For each clause, set the literal corresponding to the excluded vertex to be true.

This assignment is consistent. Indeed, suppose two excluded vertices correspond to complementary literals, such as x and $\bar{x}$. Then there is an edge between these two vertices by construction. Since both endpoints are excluded from S , this edge would not be covered, contradicting that S is a vertex cover.

Therefore, no two excluded literals are complementary, so the truth assignment is consistent.

Also, every clause is satisfied, because each clause has one excluded literal, and we set that literal to true.

Hence ϕ is satisfiable.

Therefore,

$$\phi \in 3SAT \iff \langle G, 2m \rangle \in VERTEX COVER.$$

The construction creates $3m$ vertices and polynomially many edges, so it runs in polynomial time.

Thus,

$$3SAT \leq_p VERTEX COVER.$$

Since $3SAT$ is NP-complete and $VERTEX COVER \in NP$, we conclude that $VERTEX COVER$ is NP-complete.